Dietmaier Optimization and Omnitropic descent

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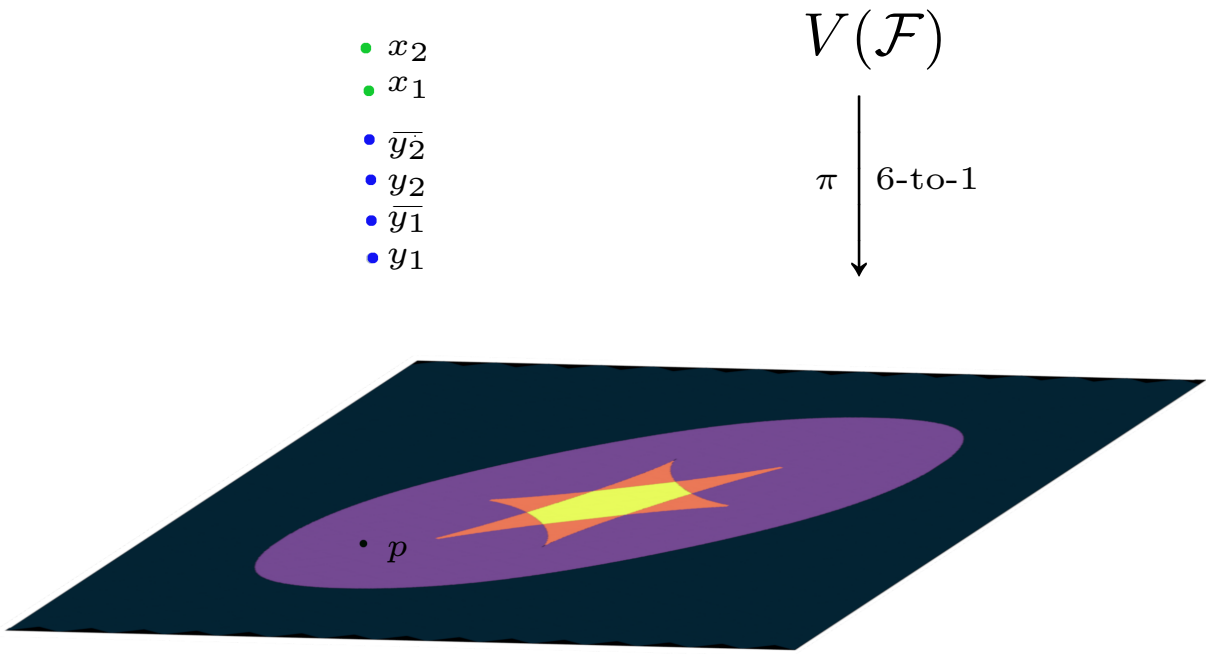
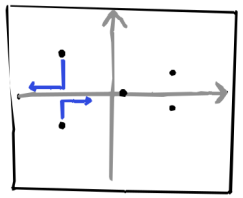
Setting and goal

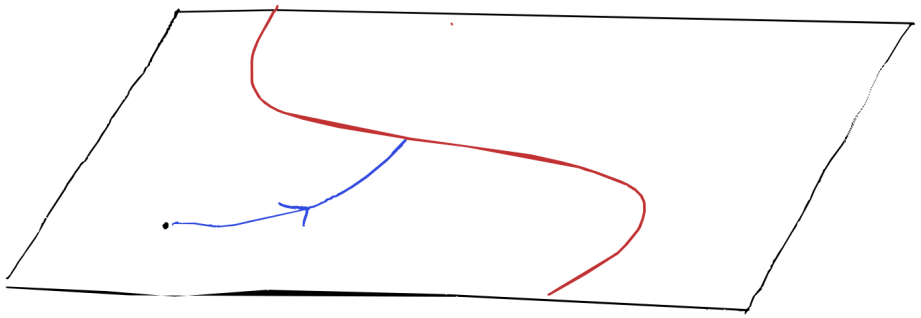
$$\mathcal{F}(z,p) = \begin{bmatrix} f_1 \\ \vdots \\ f_n \end{bmatrix} \subseteq \mathbb{C}[p_1, \cdots, p_k][x_1, \cdots, x_n]$$



Goal : To find as many real solutions as possible.

Dietmaier's idea





Dietmaier's idea: Move in $\mathbb{R}_p^k$ to smash conjugate solutions together. To do this, we need to minimize $||\text{imag } y_i(p)||$.

Dietmaier's choice

Choice: Follow gradient of $||\text{imag } y_i(p)||$ with respect to p, for y_i with the **smallest** imaginary part.
Demand: No real solutions are lost in the process.

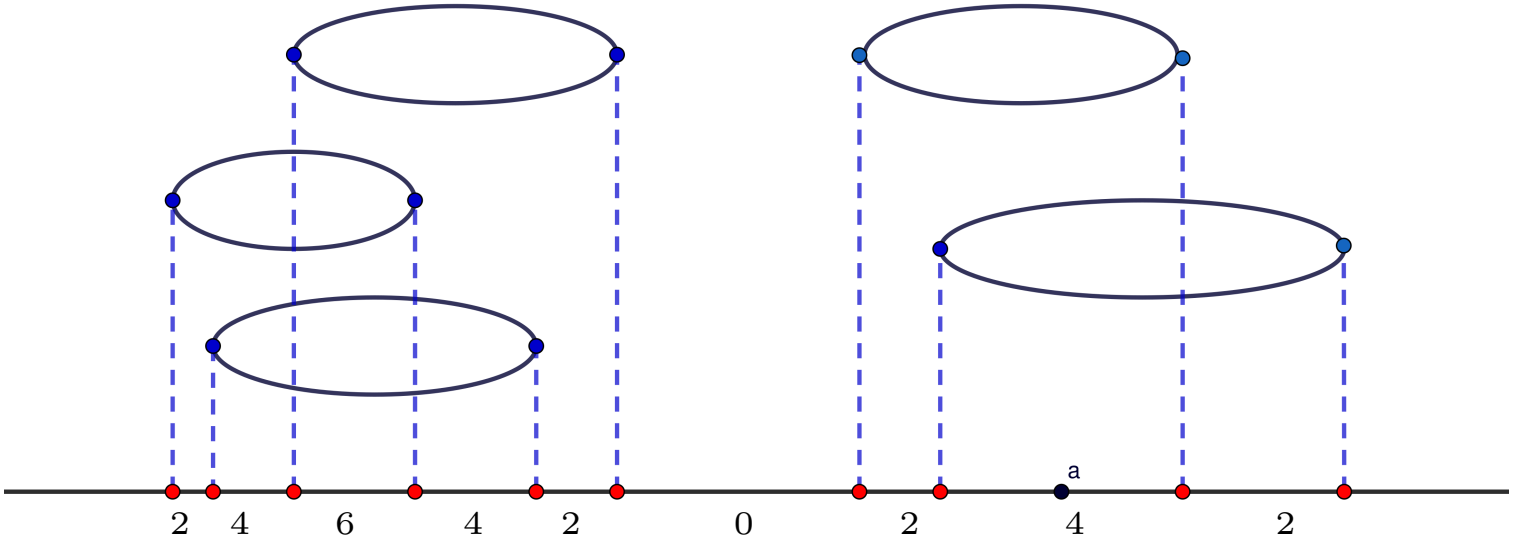
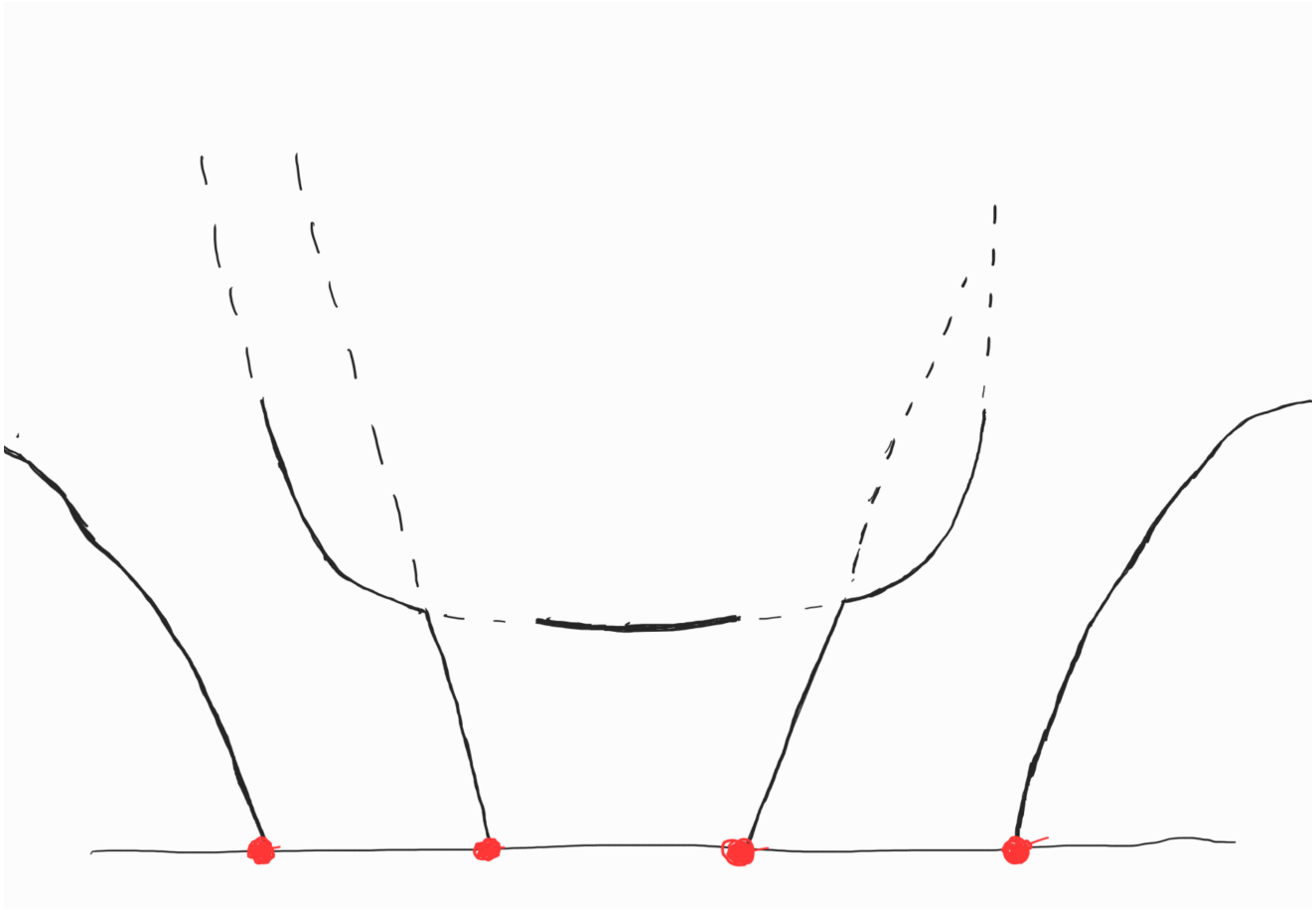
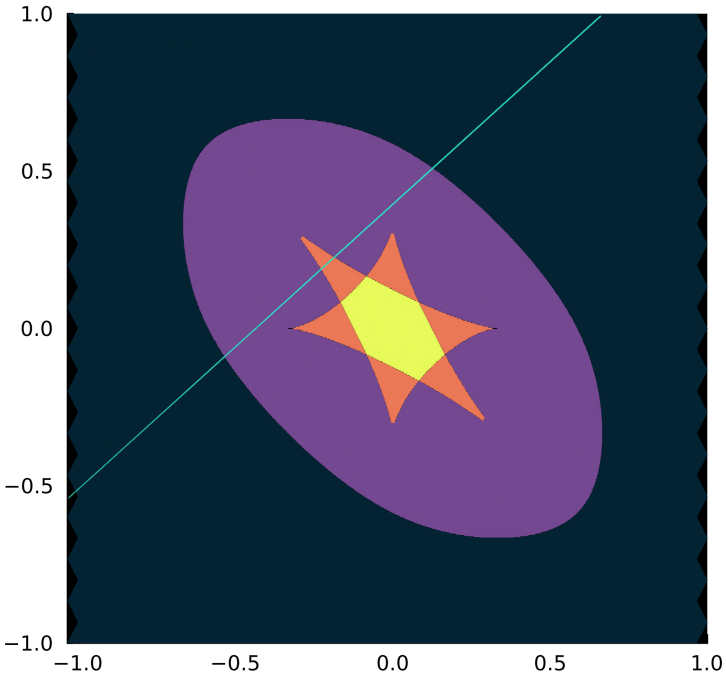
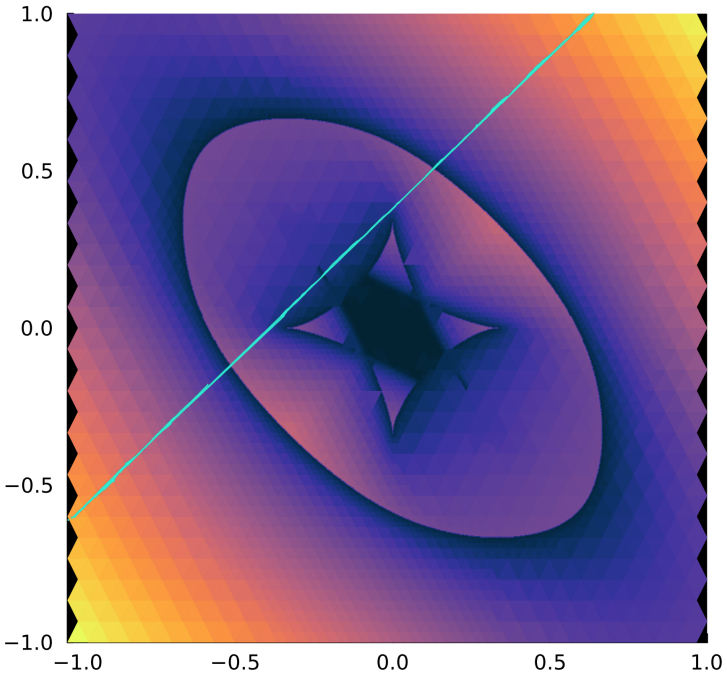


Figure 1: The demand that no real solutions are lost, will prevent us from reaching the chamber with 6 solutions.



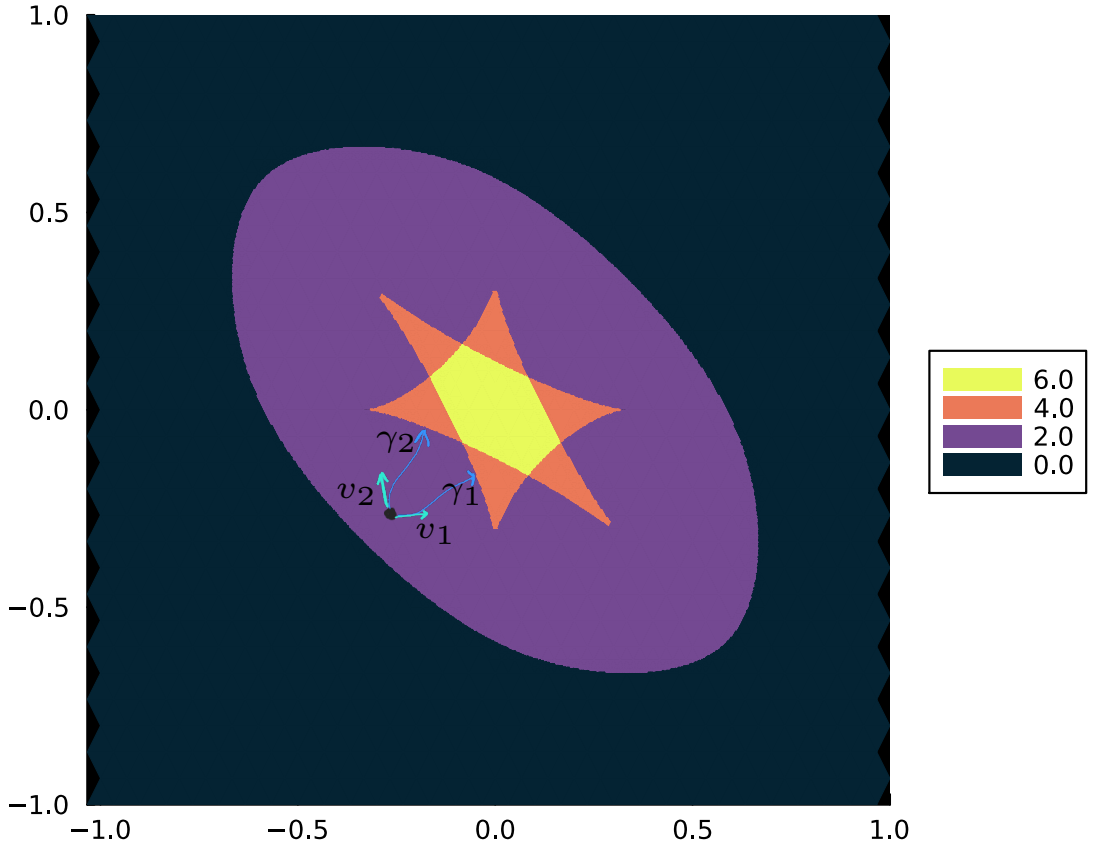


(a) Kuramoto model, restricted

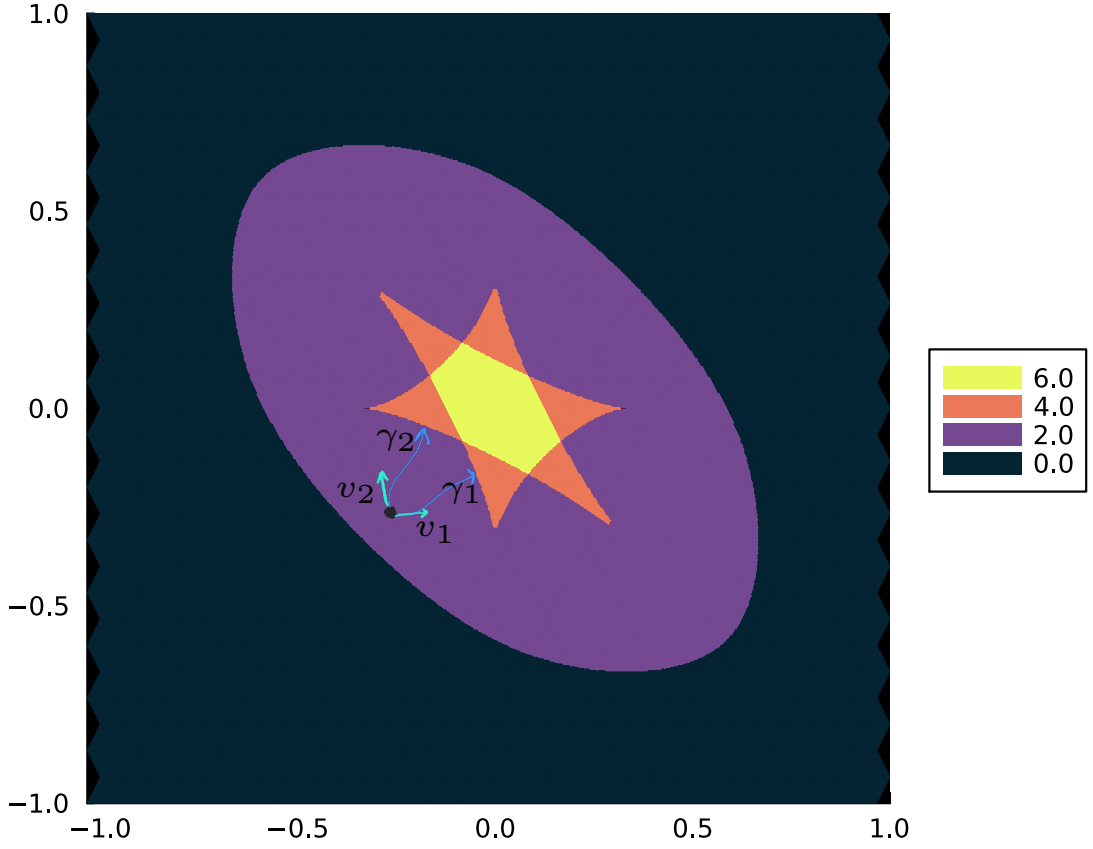


(b) The dietmaier function value

Flow directions

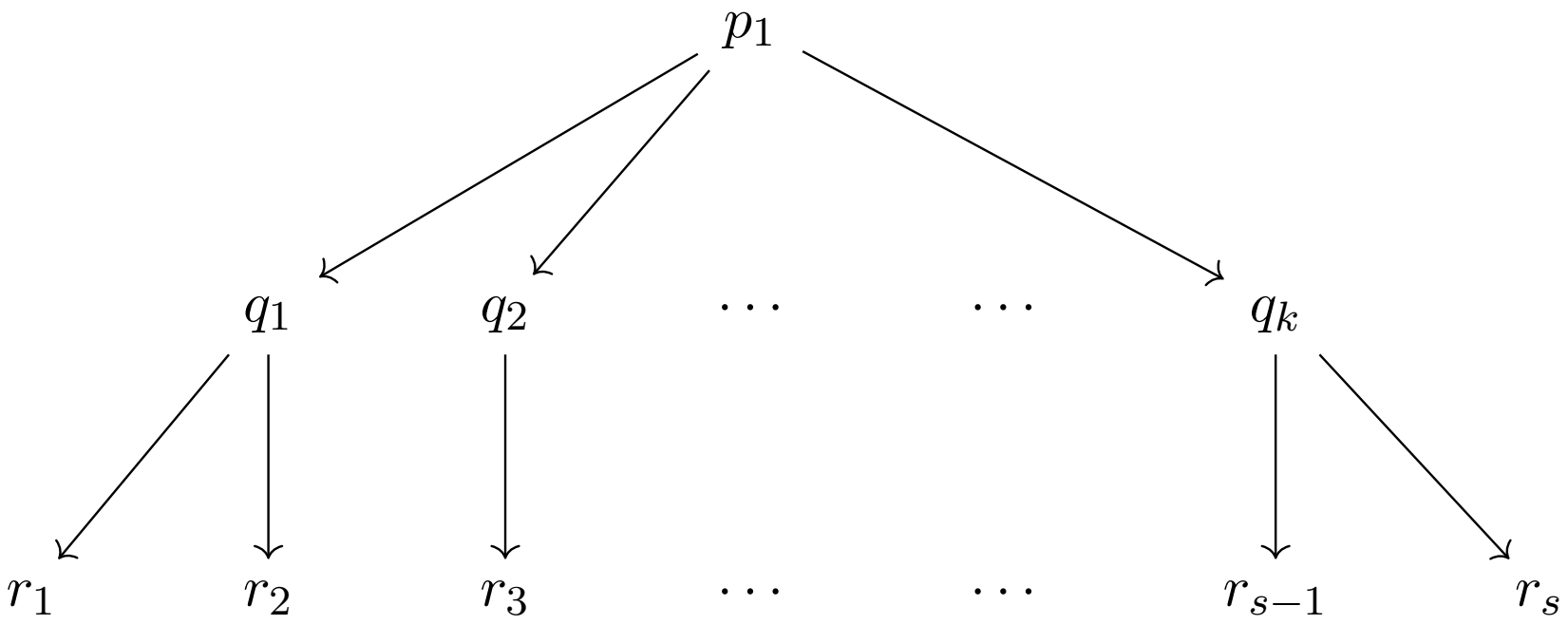


$$v_1 = -\nabla_p ||\text{imag}(y_1(p))|| \quad \gamma_1 = \text{path that takes the direction } v_1(p) \text{ at } p$$
$$v_2 = -\nabla_p ||\text{imag}(y_2(p))|| \quad \gamma_2 = \text{path that takes the direction } v_2(p) \text{ at } p.$$

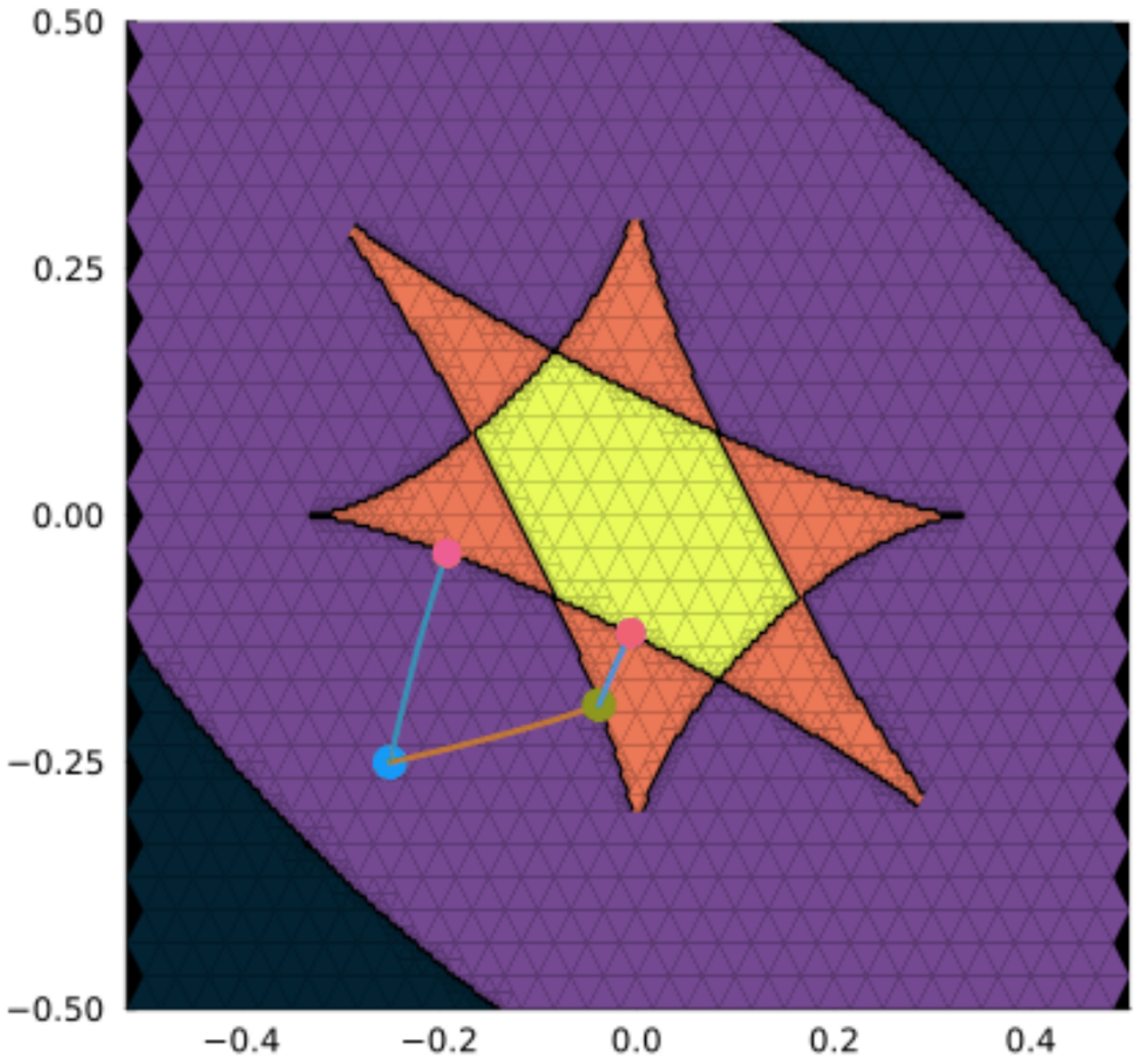


Lazy: These flows can be travelled by tracking one path, and not d paths at once. Flow tracking is very fast!
Observation: These flows need not end up in a neighbouring chamber. They might travel over the discriminant.

Omnitropic descent



Dietmaier: Chooses a branch of $y(p)$ at each step.
Omnitropic: Follow the directions given by each of the y_is lazily, and then solves for the fibres and then repeats on q_is with the most number of real solutions.


0.50
0.25
0.00
−0.25
−0.50
−0.4
−0.2
0.0
0.2
0.4

Thank You!